WEEK 2 · DAY 2 · OVERVIEW

Gradient Boosting From Residuals



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ABOUT 45 MIN WITH EXERCISES / 12 SECTIONS / 4 EXERCISES

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Problem Statement

BUSINESS CONTEXT

Course 4's Day 5 left a least squares model of log OAS that misses a held-out training bond by 41.7 bp without coupon, 37.2 with it.

Day 5 averaged trees grown side by side on the same bonds; Day 6 grew small trees one after another on the card file.

The desk asks whether trees grown in sequence, each fitted to what the others still miss, describe the bonds better on the same five folds, and what goes on inside.

The bonds and issuers are invented for this course, and a fitted spread describes them, never a fair value or a view on a bond.

OBJECTIVES

Fit gradient boosting and score it on five folds in bp, then rebuild it by hand from residuals and match the library.

Read the rounds, the learning rate and the depth from held-out curves, compare boosting with least squares and Day 5's forest without and with coupon, and find where a gain comes from.

ABOUT 45 MIN WITH EXERCISES / 12 SECTIONS / 4 EXERCISES

Your Route Through Day 2

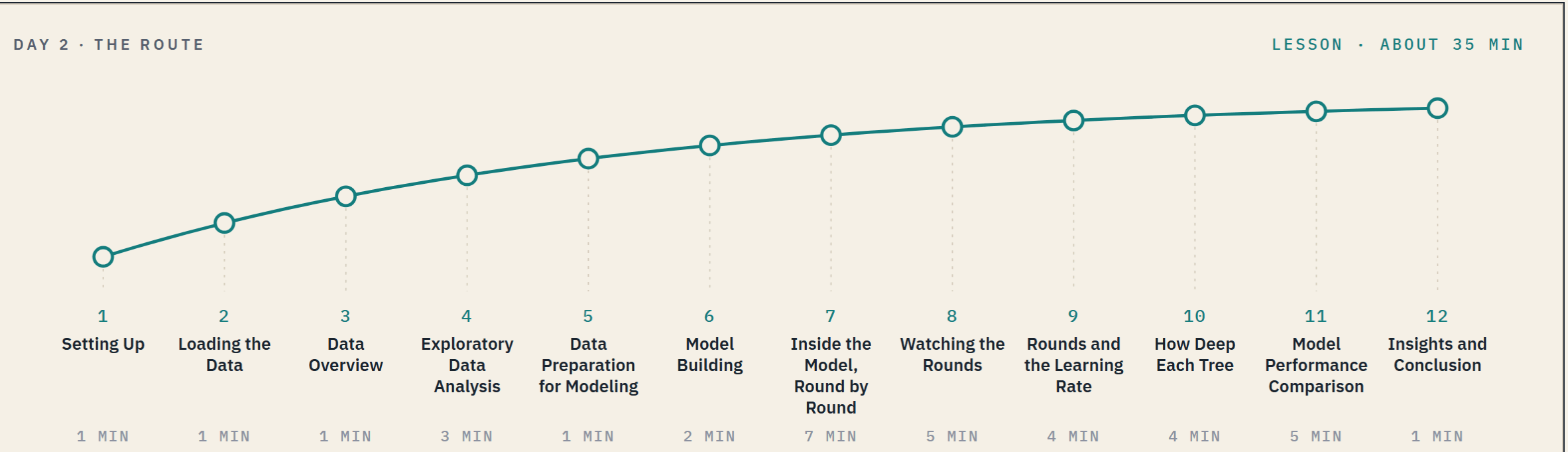


FIGURE 1 Twelve sections, in order, with about how long each takes.

Time. About 35 minutes for the lesson, then about 12 for the four exercises.

Fix the seed and load the libraries.

Setting Up

THE SETUP

```
import os
from pathlib import Path
os.environ["OMP_NUM_THREADS"] = "1"
os.environ["OPENBLAS_NUM_THREADS"] = "1"
SEED = 42
DATA = "../../../data/"
if not Path(DATA + "benchmark.csv").exists():
    DATA = "https://raw.githubusercontent.com/lenamonj/python-fixed-income/main/data/"
```

THE LIBRARIES

```
import numpy as np
import pandas as pd
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import GradientBoostingRegressor, HistGradientBoostingRegressor, RandomForestRegressor
from sklearn.metrics import mean_absolute_error, make_scorer
```

Read the bonds and keep Course 4's split.

Loading the Data

READING THE BONDS

```
bonds = pd.read_csv(DATA + "benchmark.csv")
ratings = pd.read_csv(DATA + "ratings.csv")
data = bonds.merge(ratings[["rating", "score", "grade"]], on="rating")
rating_order = list(ratings["rating"])
```

THE SPLIT

```
train, test = train_test_split(data, test_size=0.25, random_state=SEED)
y = train["log_oas"]
print("Training bonds:", len(train), " Test bonds:", len(test))
```

```
Training bonds: 615  Test bonds: 206
```

The 206 test bonds were Course 4's; Course 5 never scores them. Everything below lives inside the 615.

Data Overview

THE TRAINING BONDS

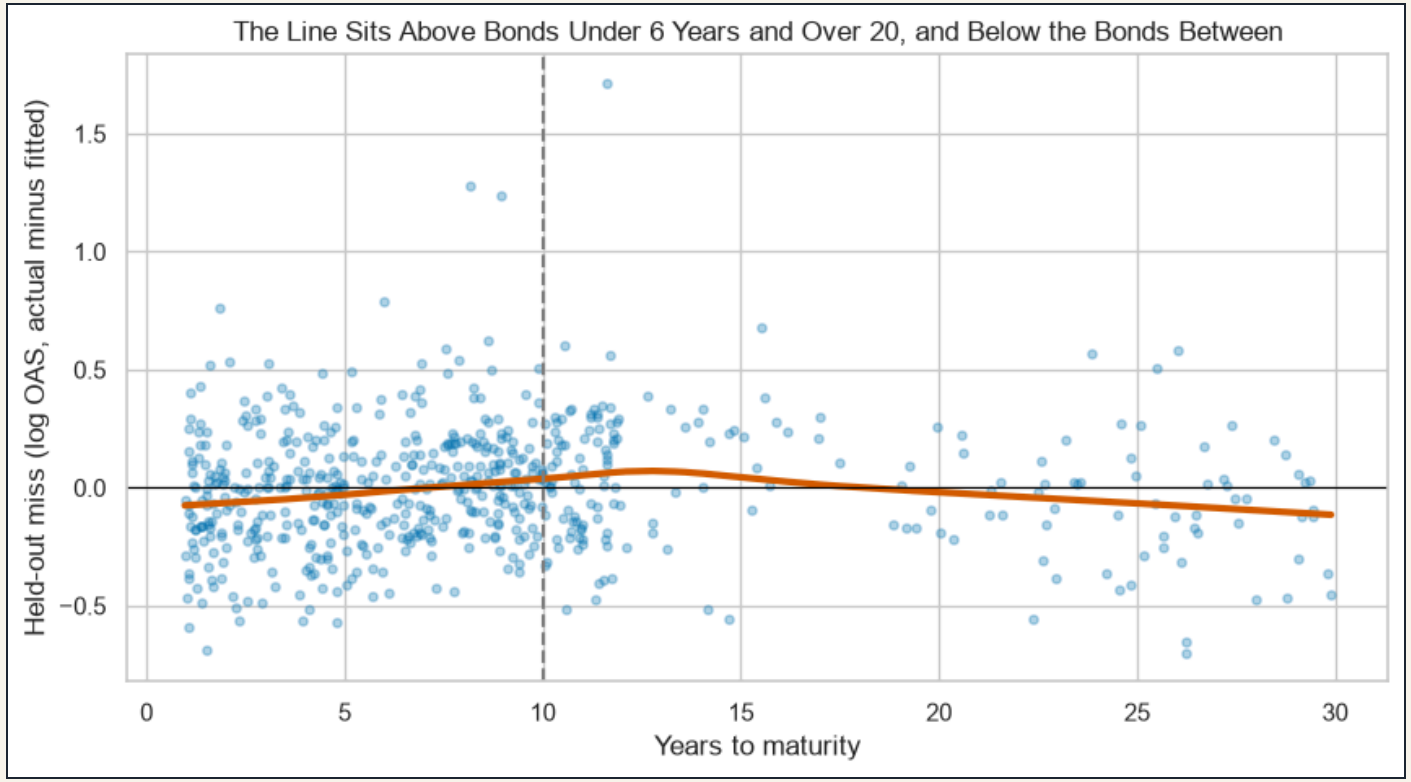
	RATING	SCORE	YEARS_TO_MATURITY	LOG_OAS
712	A+	5	4.567	3.892
181	A+	5	19.803	4.078
778	AA	3	2.538	3.091
716	BBB-	10	3.422	5.380
636	BBB+	8	19.264	4.852

THE TARGET

```
Mean OAS: 180.7 bp  Median: 123.0 bp  Widest: 2074 bp
Mean log OAS: 4.8776  exp of it: 131.3 bp
```

Familiar bonds and a familiar target: what we need is Course 4's best line to beat.

Exploratory Data Analysis



COURSE 4'S LINE, FIVE FOLDS

```
Folds: [30.1 36.2 61.8 39.3 41.2]  Mean: 41.72 bp
```

THE MISS BY MATURITY

YEARS_TO_MATURITY	SIZE	MEAN
(0, 2]	74	-0.064
(2, 4]	78	-0.017
(4, 6]	93	-0.058
(6, 8]	81	0.018
(8, 10]	108	0.061
(10, 12]	84	0.058

A straight line in years cannot bend at 10. Next, a model that finds its own bends.

Build 14 columns for the trees.

Data Preparation for Modeling

THE TREE INPUTS

```
(615, 14)
```

WHY NOTHING IS SCALED

A tree splits a number as it stands, so nothing is scaled; indicator columns learn nothing from the rows, so making them before the folds leaks nothing.

WHAT THE FILE HOLDS

- **`cusip`**: the bond's identifier
- **`issuer`** and **`ticker`**: the issuer's name and short code
- **`sector`**: one of 10 industry sectors
- **`rating`**: credit rating, AAA to CCC
- **`coupon`**: annual coupon, percent of par
- **`maturity`** and **`as_of_date`**: dates; years to maturity is built from them
- **`oas`**: option-adjusted spread, bp; **the target is its log**
- **`duration`**: modified duration, years
- **`amount_outstanding`**: face value outstanding, dollars

Same bonds, same folds, same miss in bp: only the model changes.

Fit boosting with its defaults and score it.

Model Building

THE DEFAULTS

```
gbr = GradientBoostingRegressor(random_state=SEED)
```

```
Rounds: 100  Learning rate: 0.1  Depth of each tree: 3
```

WHAT BOOSTING DOES

It starts every bond at the average log OAS, fits a small tree to what is still missed, adds a tenth of that tree's output, and repeats.

SCORED ON THE SAME FIVE FOLDS

```
scores = cross_validate(gbr, X, y, cv=folds, scoring=bp_scorer, return_train_score=True, n_jobs=1)
```

```
Held-out folds: [27.2 36.9 49.2 33.9 37.2]  Mean: 36.88 bp
Fitted folds, mean: 23.45 bp
```

Three lines beat Course 4's best line by 4.8 bp. Next we open the 100 trees.

Inside the Model, Round by Round

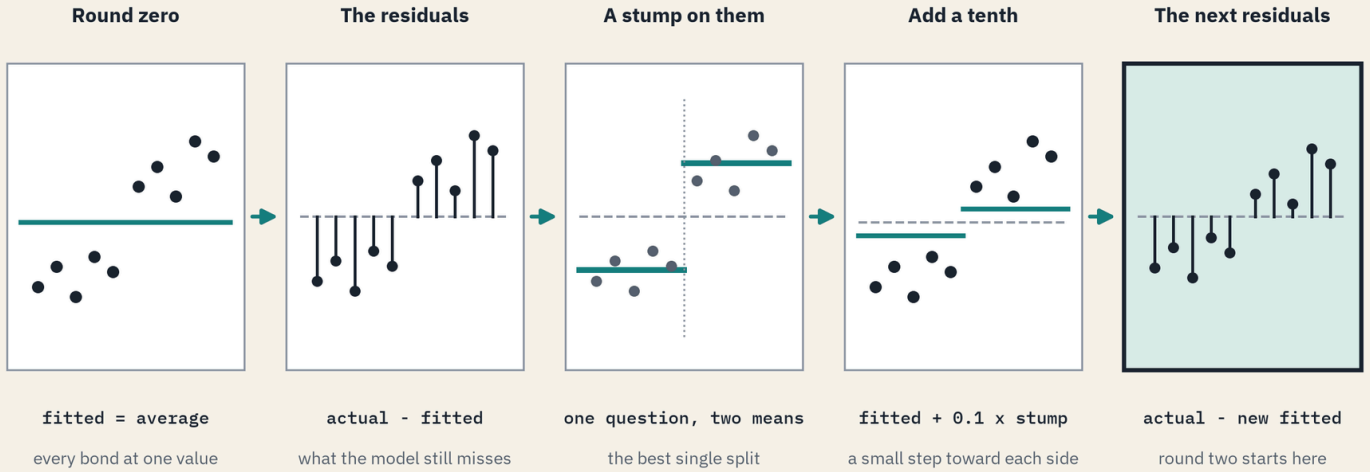


FIGURE 2 One round: every bond at the average, the residuals, a stump on them, a tenth added, and the next residuals.

ROUND ZERO

```
fitted0 = np.full(len(y), y.mean())
residuals0 = y - fitted0

Every bond at 131.3 bp
Squared residuals: 354.13
```

THE QUESTION

Q. One question splits the bonds in two, and each side is fitted at its own average residual. What percent of the 354.13 squared residuals can the best single question remove? Write down a number.

Gradient boosting is the mean plus a hundred small trees, each fitted to what is left, each added at a tenth.

Step through the rounds and follow three bonds.

Watching the Rounds

STUMPS ON SCORE ALONE

```
score_model = GradientBoostingRegressor(n_estimators=300, max_depth=1, random_state=SEED).fit(train[["score"]], y)
```

YOUR TURN: THE SLIDER

```
def show_round(round_number=1):
interact(show_round, round_number=(1, 300));
```

THE QUESTION

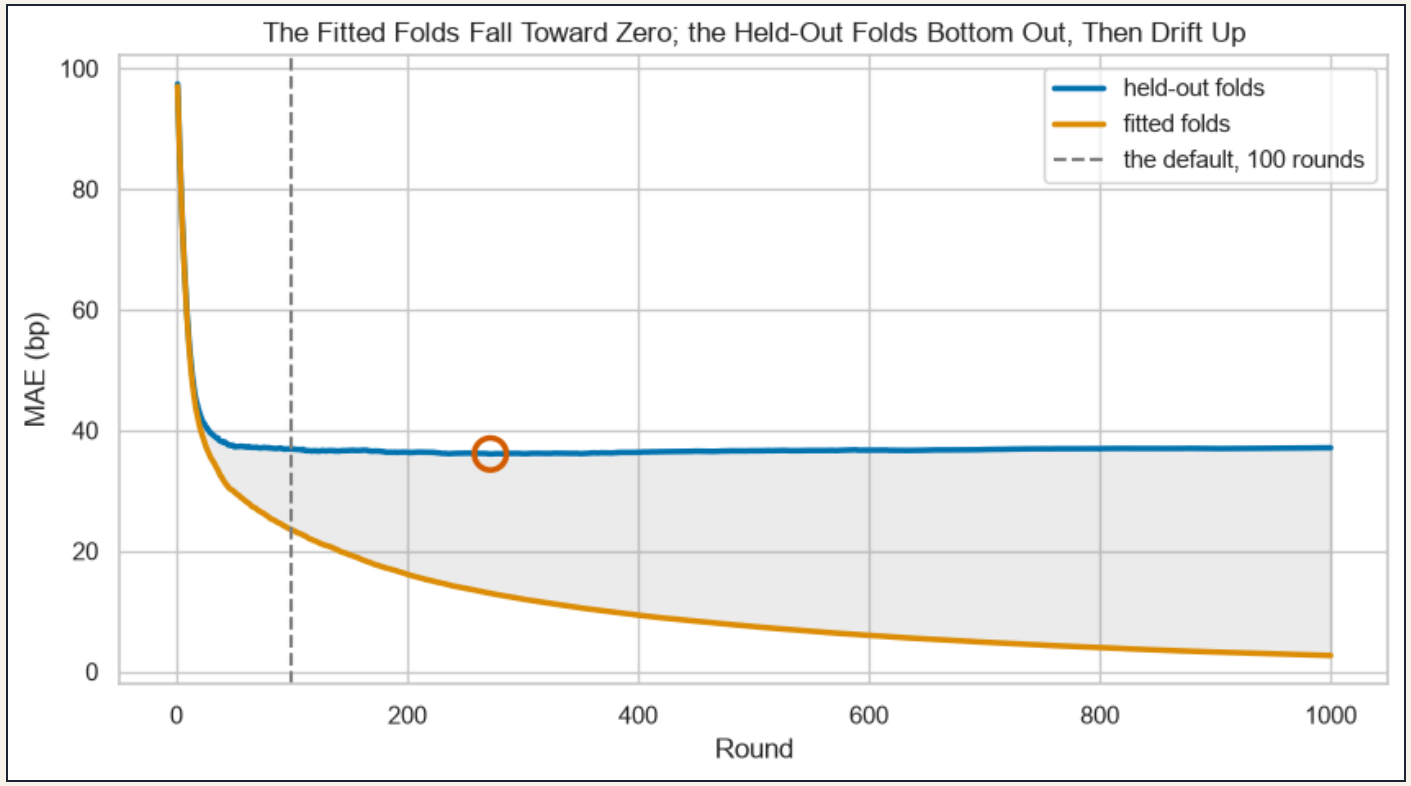
Q. Sketch the fitted spread against rating score after 300 rounds.

THREE BONDS TO FOLLOW

	CUSIP	ISSUER	RATING	OAS
158	99030EAG0	Dralane Networks	AAA	15
449	99085HAB8	Tromazor Resources	BBB-	131
386	99072UAF5	Zephane Utilities	B	2074

Every round closes on the bonds it sees, the widest one included. That is how boosting memorises, so we watch rounds on unseen bonds.

Rounds and the Learning Rate



ONE RATE, 1,000 ROUNDS

```
curves = {0.1: round_curves(0.1)}
best_round = curves[0.1]["held-out"].idxmin()

Lowest held-out MAE: 36.1 bp at round 272
```

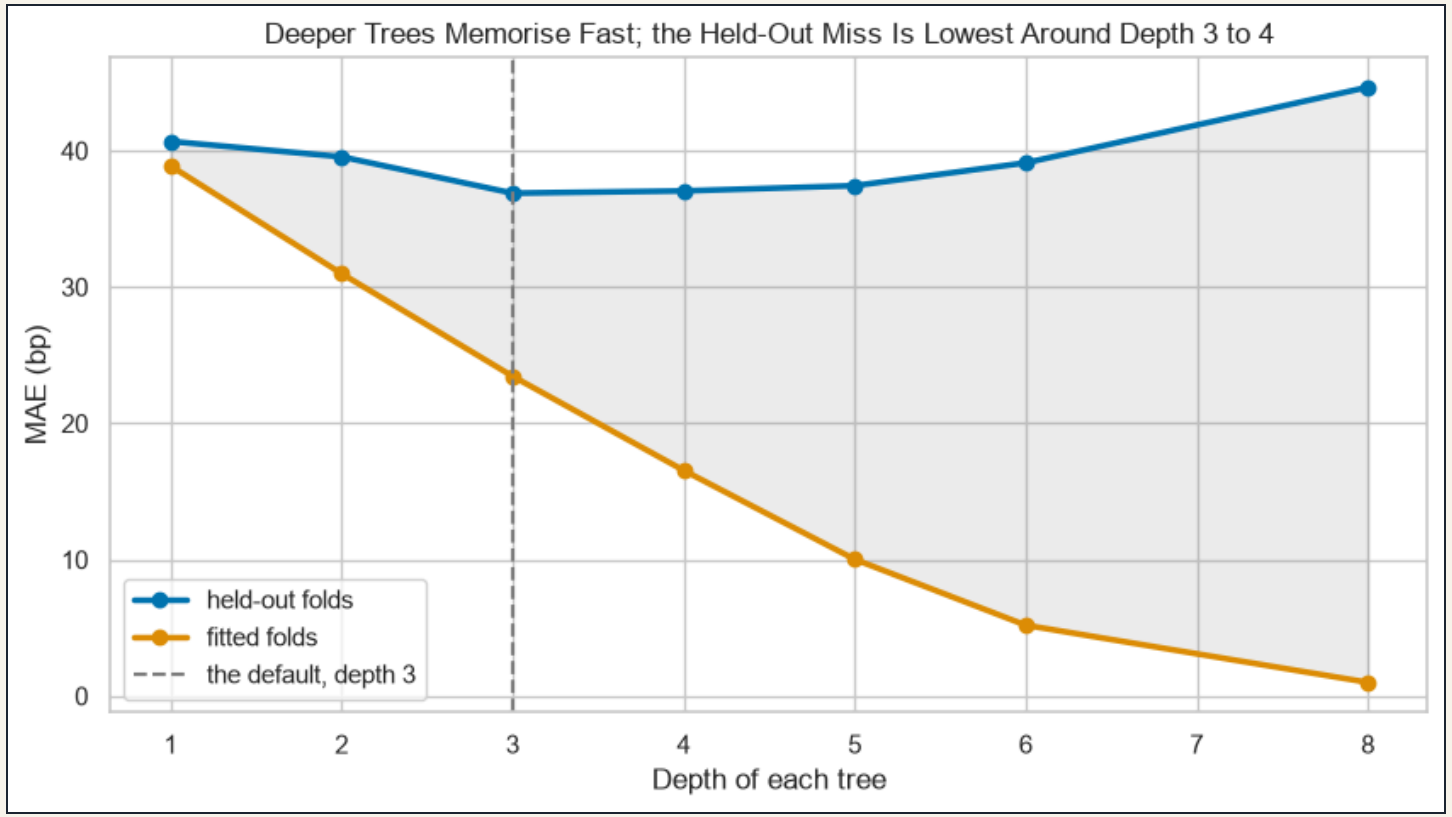
WHAT THE CURVES SHOW

The fitted-fold miss falls from 23.5 bp at round 100 to 2.7 at round 1,000: the model memorises its rows.

The held-out miss is lowest at round 272, 36.1 bp, and drifts up to 37.1 by round 1,000.

These curves describe the five folds, and each lowest point flatters the setting that drew it. We keep the defaults, fixed before looking; Day 11 tunes honestly.

How Deep Each Tree

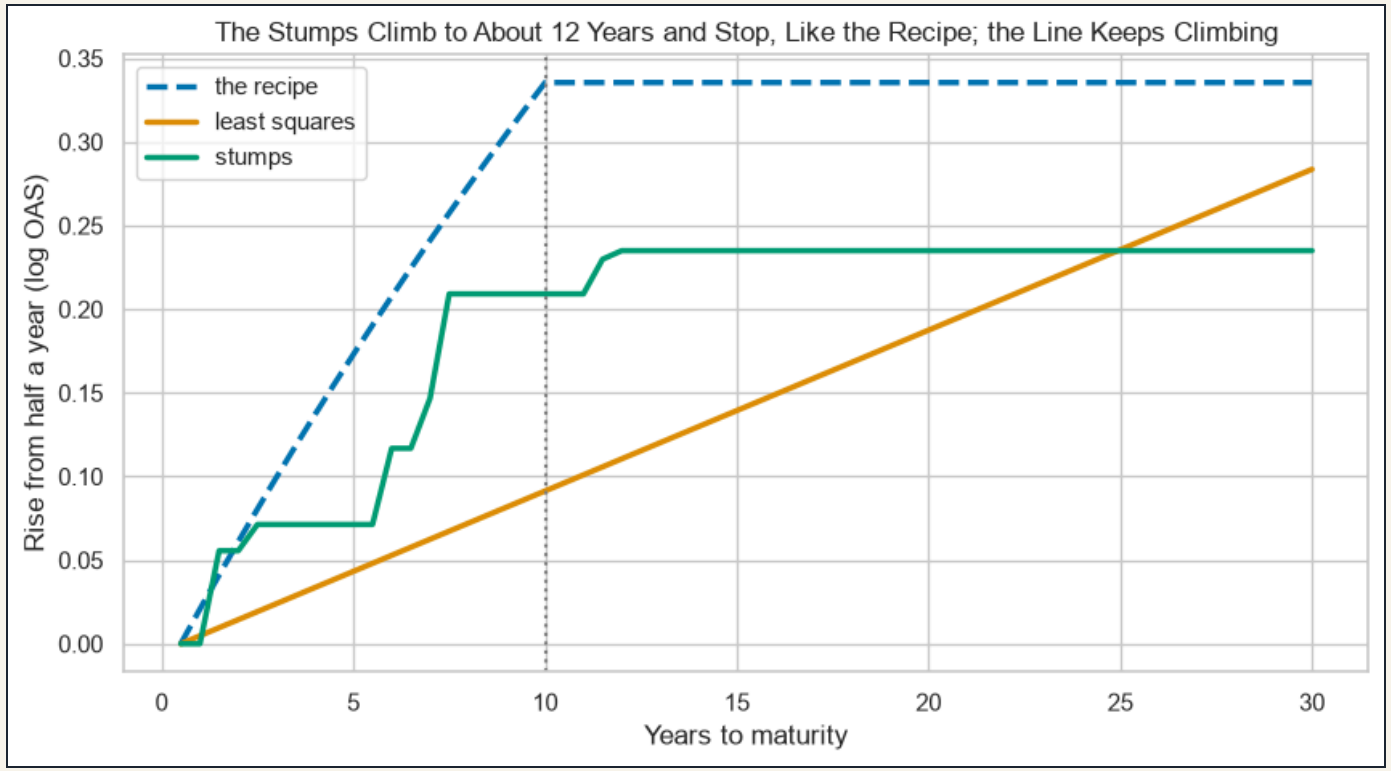


DEPTH 1 TO 8, FIVE FOLDS

	FITTED FOLDS	HELD-OUT
1	38.884	40.680
2	30.982	39.552
3	23.451	36.880
4	16.549	37.045
5	10.023	37.430
6	5.191	39.105
8	0.997	44.651

Depth lets a tree read years and duration together, where this file hides part of the spread. Stumps cannot.

Model Performance Comparison



THE SHAPES AT 10, 20 AND 30 YEARS

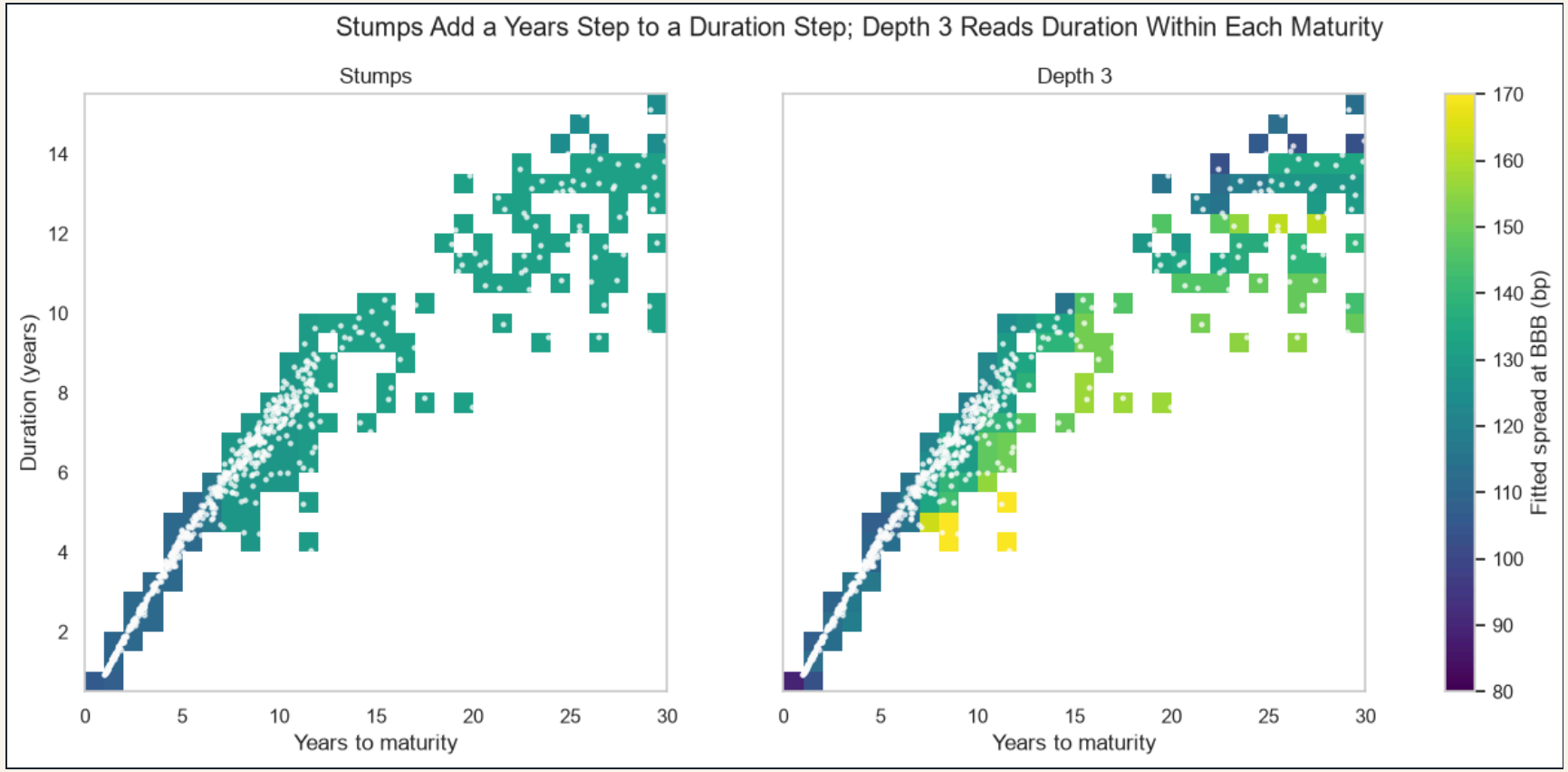
	THE RECIPE	LEAST SQUARES	STUMPS
10.000	0.335	0.091	0.209
20.000	0.335	0.187	0.235
30.000	0.335	0.284	0.235

WHERE BOOSTING MISSES

```
boosting   line
grade
High Yield          72.092 85.886
Investment Grade    18.489 18.648
```

High yield bonds carry the misses: 72.1 bp for boosting against 18.5 for investment grade, and nearly all the gain on the line (85.9 and 18.6) is in high yield.

Insights and Conclusion



Day 8 runs the same rounds on log-odds for the card file, with XGBoost's penalties and early stopping. `GradientBoostingClassifier` **is scikit-learn's version of that.**

You Can Now

- Fit **`GradientBoostingRegressor`** and score it on folds in bp, held-out and fitted.
- Run a split search by hand and match a stump's question with **`.tree_`**.
- Rebuild gradient boosting as a loop over residuals and match the library.
- Follow a model round by round with **`.staged_predict`**.
- Read rounds, learning rate and depth from held-out curves, and name an interaction.

NEXT
WEEK 2 · DAY 3

XGBoost, Regularization, and Early Stopping

Day 8 runs the same rounds on log-odds for the card file, with XGBoost.